BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 025 · Network Topologies

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does DNS do?

Q2. Why is a thermostat controller an embedded system?
A. It has a specific task
B. It runs any desktop app
C. It always needs Wi-Fi
D. It always has a screen

Q3. What makes a system embedded?

Q4. A circuit uses two voltage ranges for two states. “Computers use binary because people cannot write numbers above 1.” Correct this. Why are two states useful?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Topology — Arrangement of network devices and links.

Star topology — Endpoints connect to a central switch or hub.

Mesh topology — Devices connect via multiple paths between nodes.

Fault tolerance — Ability to continue working after some faults.

Central switch — Network device that connects and forwards data to LAN devices.

Redundancy — Extra paths that can be used if a path fails.

New learning

Topology describes connections: A network topology shows how computers and connection devices link together.

Star topology: Each endpoint connects to a central device such as a switch. Easier to add/manage devices, but the central device is a point of failure.

Mesh topology: Devices can connect to multiple others. Alternative paths make the system resilient if one link fails, but more connections increase cost and complexity.

Compare in context: For a small school LAN, a star can be practical. A high-availability network may use mesh routes; explain what happens when a link or central device fails.

Worked example

A school LAN uses a star topology. Explain one advantage and one disadvantage if its central switch fails. [4 marks]

1. All classroom devices connect to one central switch.

2. New devices can be connected easily [1], making the network straightforward to manage [1].

3. The switch is a single point of failure [1], so its failure can interrupt communication for connected devices [1].

4. Give advantage plus linked benefit and disadvantage plus linked effect.

We Do · Practise with guidance

Guided 1

A business needs reliable connections between several sites. Explain one advantage and one disadvantage of a mesh topology. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Q3. Star devices connect to a central switch [1], are easier/cheaper to manage [1], but depend on the central device [1]. Mesh provides more than one path [1], so it can withstand some link failures [1], but uses extra connections increasing cost/complexity [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Distributed DNS servers help look up the IP address associated with a domain name.

Q2. A — It has a specific task

Q3. It is built into a larger device for a specific task, often with limited resources.

Q4. Circuits can reliably tell two states apart: 0 and 1. These states represent all kinds of data in binary.

I Do / We Do pages · Guided practice

Q1. Multiple possible routes [1] mean traffic can use another path if a link fails [1]. More connections/hardware are needed [1], increasing cost or maintenance [1].

Q2. Each device has its own link to the central device [1], making fault isolation easier [1]. Devices can be added or managed centrally [1], simplifying expansion [1].

Q3. Mesh provides multiple paths between devices [1]. When a link fails [1], traffic may be routed along another available path [1], keeping communication possible [1].

You Do pages · Code First

```
devices = 4
links = devices * (devices - 1) DIV 2
print(links)
```

Expected result: 6

You Do pages · Independent questions

Q1. Several routes provide fault tolerance if a link fails [1].

Q2. The central switch connects the kiosks/computers [1], so its failure can interrupt network communication [1]. A spare or quickly replaceable switch, or a resilient design, can reduce the problem [1] by restoring service/reducing downtime [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Worked method: Multiple possible routes [1] mean traffic can use another path if a link fails [1]. More connections/hardware are needed [1], increasing cost or maintenance [1].

Guided 2

A small primary school chooses a star topology. Explain two advantages of this decision. [4 marks]

Worked method: Each device has its own link to the central device [1], making fault isolation easier [1]. Devices can be added or managed centrally [1], simplifying expansion [1].

Guided 3

Describe how a mesh can continue operating after one connection fails. [4 marks]

Worked method: Mesh provides multiple paths between devices [1]. When a link fails [1], traffic may be routed along another available path [1], keeping communication possible [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the count of links for a small full mesh.

```
devices = 4
links = devices * (devices - 1) DIV 2
print(links)
```

predict

Question 1

State one advantage of a mesh topology. [1 mark] Source: Adapted from your Paper 1 revision booklet, 1.3.1.

Question 2

A cinema uses a star network to connect ticket kiosks and staff computers. The central switch stops working. Explain the effect and suggest one precaution. [4 marks] Source: Original J277-style question, not an OCR past-paper original.

Question 3

Compare star and mesh topologies for a large organisation. [6 marks] Source: Original J277-style question, not an OCR past-paper original.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

Mesh topology generally provides alternative links that can improve fault tolerance.